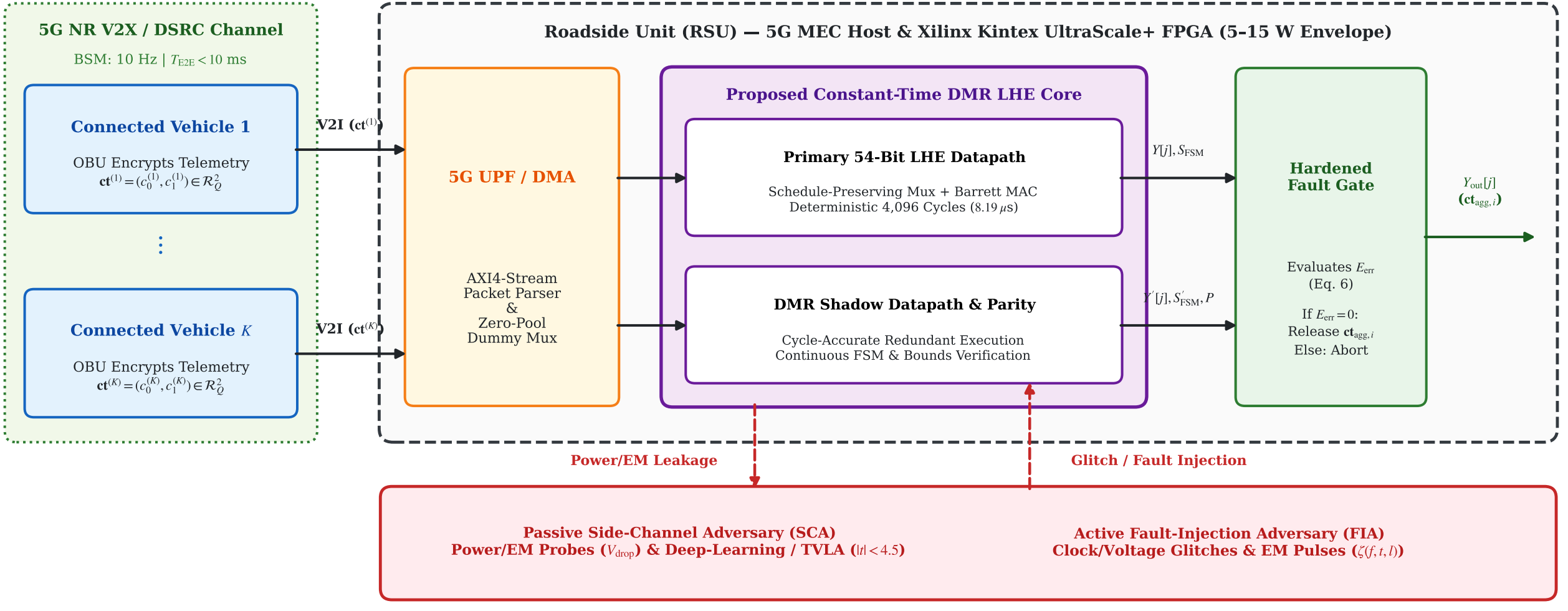

5G NR V2X / DSRC Channel
BSM: 10 Hz | $T_{E2E} < 10$ ms
Connected Vehicle 1
OBU Encrypts Telemetry
$\mathbf{ct}^{(1)} = (c_0^{(1)}, c_1^{(1)}) \in \mathcal{R}_Q^2$
Connected Vehicle K
OBU Encrypts Telemetry
$\mathbf{ct}^{(K)} = (c_0^{(K)}, c_1^{(K)}) \in \mathcal{R}_Q^2$
V2I ($\mathbf{ct}^{(1)}$)
V2I ($\mathbf{ct}^{(K)}$)
Roadside Unit (RSU) — 5G MEC Host & Xilinx Kintex UltraScale+ FPGA (5–15 W Envelope)
5G UPF / DMA
AXI4-Stream
Packet Parser
&
Zero-Pool
Dummy Mux
Proposed Constant-Time DMR LHE Core
Primary 54-Bit LHE Datapath
Schedule-Preserving Mux + Barrett MAC
Deterministic 4,096 Cycles (8.19 μs)
DMR Shadow Datapath & Parity
Cycle-Accurate Redundant Execution
Continuous FSM & Bounds Verification
$Y[j], S_{FSM}$
$Y'[j], S'_{FSM}, P$
Hardened
Fault Gate
Evaluates E_{err}
(Eq. 6)
If $E_{err} = 0$:
Release $\mathbf{ct}_{agg,i}$
Else: Abort
$Y_{out}[j]$
($\mathbf{ct}_{agg,i}$)
Power/EM Leakage
Glitch / Fault Injection
Passive Side-Channel Adversary (SCA)
Power/EM Probes (V_{drop}) & Deep-Learning / TVLA ($|t| < 4.5$)
Active Fault-Injection Adversary (FIA)
Clock/Voltage Glitches & EM Pulses ($\zeta(f, t, l)$)